

Hopfield Networks is All You Need

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Understanding Transformer Attention

Through the Lens of Associative Memory



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고전 홉필드

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결론

"Hopfield networks or associative memories store and retrieve patterns."

Classical Hopfield Networks

고전 홉필드 네트워크

"Hopfield networks or associative memories store and retrieve patterns." — Ramsauer et al.

Store patterns

Retrieve with noise

Energy minimization

What is a Hopfield Network? / 홉필드 네트워크란?

PART 1

Concept / 개념

A neural network that can store a set of patterns as "memories" and retrieve the closest one given a noisy or partial input.

손상된 이미지 → 원본 복원
노이즈 섞인 패턴 → 가장 가까운 기억으로 수렴

Setup / 기본 구성

State vector $x \in \{-1, +1\}^d$

Stored patterns $\xi^\mu, \mu=1..N$

N : number of stored patterns

d : number of neurons

Key property: the network always converges to a stored pattern (energy minimum).

Energy Function / 에너지 함수

$$E = -\frac{1}{2} \mathbf{x}^T W \mathbf{x}$$

Key: Update steps always decrease E

→ Network converges to local minimum

→ Local minima = stored patterns

Weight Matrix (Hebb's Rule) / 가중치 행렬

$$W = \frac{1}{N} \sum_{\mu=1}^N \xi^{\mu} (\xi^{\mu})^T$$

Interpretation: Neurons that fire together, wire together (Hebb, 1949)

Update rule / 업데이트 규칙

$$x_i \leftarrow \text{sign}\left(\sum_j w_{ij} x_j\right)$$

Storage Capacity

$$C_{\text{classical}} \approx 0.14 \times d$$

$d = 1,000$ neurons $\rightarrow$ only ~140 patterns

Four Key Limitations / 4가지 한계

Linear capacity

Only $0.14d$ patterns — far too small

Binary states only

Cannot handle continuous real-valued data

Slow convergence

Requires many update iterations

Spurious states

Converges to unintended 'ghost' memories

Dense Associative Memory

조밀 연상 기억

"Dense Associative Memories allow for an exponential number of patterns to be stored." — Krotov & Hopfield (2016)

Nonlinear interaction F

Polynomial capacity $d^{(n-1)}$

Classical / 기존 (n=2)

$$F(z) = z^2 \text{ (quadratic)}$$

$$E = -\frac{1}{2} \mathbf{x}^T W \mathbf{x}$$

Capacity: $C \sim 0.14d$ (linear)

Only simple quadratic interactions

Dense Associative / 조밀 (n >= 2)

$$E = -\sum_{\mu} F\left(\sum_i x_i \xi_i^{\mu}\right) + \frac{1}{2} \sum_i x_i^2$$

Polynomial $F(z) = z^n$: when $n > 2$, interactions become higher-order

Capacity grows as **$d^{(n-1)}$** — polynomial in d

$$F(z) = z^n, \quad n \geq 2$$

Capacity Scaling with Interaction Order n

PART 2

Model	Interaction $F(z)$	Capacity	$d=1000$
Classical ($n=2$)	z^2	$\sim 0.14d$	~ 140
Dense $n=3$	z^3	$\sim d^2$	$\sim 1,000,000$
Dense $n=4$	z^4	$\sim d^3$	$\sim 10^9$
Dense $n=\infty$ $\rightarrow \exp(z)$	$\exp(z)$ (softmax limit)	$2^{(d/2)}$	$2^{500} \approx \infty$

"We show that the special case of the softmax function as interaction function connects the Hopfield network to the attention mechanism of transformer networks."

As $n \rightarrow \infty$ (exponential F), capacity becomes exponential — this is the key insight connecting to softmax

Modern Hopfield Networks with Continuous States

연속 상태를 가진 현대 홉필드 네트워크

"We generalize Hopfield networks to continuous states." — Ramsauer et al.

Continuous states

New energy function

Update rule derivation

Classical / 이진 상태

State: $x \in \{-1, +1\}^d$

Patterns: $\xi^\mu \in \{-1, +1\}^d$

Limitation: cannot represent
gradient or continuous data

Modern / 연속 상태

Query state: $\xi \in \mathbb{R}^d$

Stored patterns: $X = [x^1 \dots x^N] \in \mathbb{R}^{(d \times N)}$

Interaction: $F(z) = \exp(z)$

($\rightarrow$ softmax normalization)

Result: exponential storage capacity

Energy function / 에너지 함수 (Eq. 6 in paper)

$$E = -\text{lse}(\beta, X^T \xi) + \frac{1}{2} \xi^T \xi + \beta^{-1} \ln N + \frac{1}{2} M_{\max}^2$$

log-sum-exp (lse) / 정의

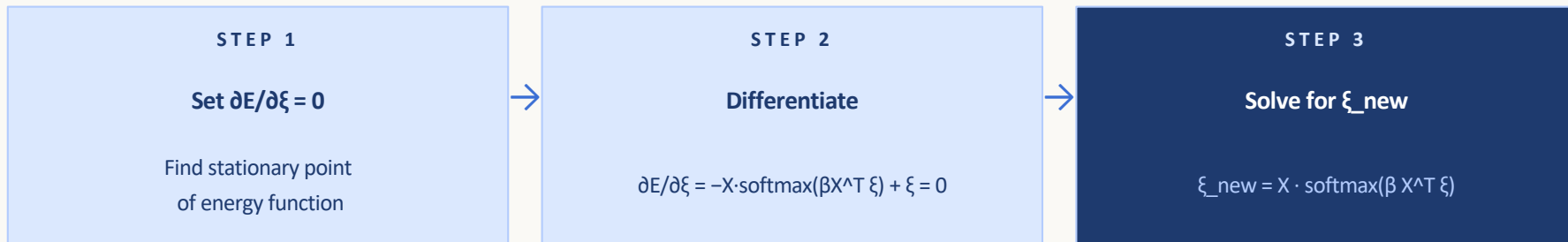
$$\text{lse}(\beta, \mathbf{z}) = \beta^{-1} \ln \left(\sum_{i=1}^N e^{\beta z_i} \right)$$

Term-by-term / 항별 해석

$-\text{lse}(\beta, X^T \xi)$ attracts ξ to nearest stored pattern

$\frac{1}{2} \|\xi\|^2$ regularization (keeps ξ bounded)

$\beta^{-1} \ln N$ constant offset



Gradient / 에너지 미분

$$\frac{\partial E}{\partial \xi} = -X \cdot \text{softmax}(\beta X^T \xi) + \xi = 0$$

Update Rule / 업데이트 규칙

$$\xi_{new} = X \cdot \text{softmax}(\beta X^T \xi)$$

Setting $\partial E / \partial \xi = 0$ yields the softmax-weighted centroid — the core of Transformer attention

$\beta \rightarrow \infty$ (sharp focus)

$$\beta \rightarrow \infty \Rightarrow \text{softmax}(\cdot) \rightarrow \text{one-hot}(\text{argmax})$$

Softmax concentrates on the single most similar stored pattern.

Equivalent to nearest-neighbor retrieval.

집중 검색: 가장 유사한 패턴 하나에 집중

$\beta \rightarrow 0$ (broad average)

$$\beta \rightarrow 0 \Rightarrow \text{softmax}(\cdot) \rightarrow \frac{1}{N} \mathbf{1} \text{ (uniform)}$$

Softmax becomes uniform — output is the average of all stored patterns.

No focused retrieval; all patterns contribute equally.

균등 검색: 모든 패턴 동등 혼합

In Transformer: $\beta = 1/\sqrt{d_k}$ — automatically calibrates focus based on embedding dimension d_k

Transformer Attention as Modern Hopfield Network

트랜스포머 어텐션과의 동치성

"The special case of softmax as interaction function connects the Hopfield network to the attention mechanism." — Ramsauer et al.

Mathematical equivalence

Component mapping

Multi-Head interpretation

Two Formulas — One Operation / 두 수식, 동일한 연산

PART 4

Modern Hopfield Update Rule

현대 홉필드 업데이트 규칙

Ramsauer et al., ICLR 2021

$$\xi_{new} = X \cdot \text{softmax}(\beta X^T \xi)$$

X stored pattern matrix

ξ current query state

β inverse temperature

softmax weighted average

Transformer Attention

트랜스포머 어텐션

Vaswani et al., NeurIPS 2017

$$\text{Attention}(Q, K, V) = V \cdot \text{softmax}\left(\frac{K^T Q}{\sqrt{d_k}}\right)$$

V Value matrix

K Key matrix

Q Query vector

$1/\sqrt{d_k}$ scale factor

=

Setting $\beta = 1/\sqrt{d_k}$ makes the two formulas mathematically identical — Transformer attention IS Hopfield retrieval

Hopfield	Role	Transformer
ξ (query state)	What to search for?	Q (Query)
X^K (key patterns)	What is available?	K (Key)
X^V (value patterns)	What to return?	V (Value)
β (inverse temperature)	How sharply to focus?	$1 / \sqrt{d_k}$
N (pattern count)	How many memories?	Sequence length
$\text{softmax}(\beta X^T \xi)$	Attention weights	$\text{softmax}(K^T Q / \sqrt{d_k})$

Every component in Transformer attention has a direct Hopfield counterpart — the architectures are structurally identical

Multi-Head Attention as Parallel Hopfield Networks

PART 4

"Multiple heads correspond to multiple Hopfield networks running in parallel, each attending to different patterns."

Head 1	Syntactic relations	문법적 의존 관계
Head 2	Semantic similarity	의미적 유사도
Head 3	Positional context	위치 및 순서 관계
Head 4	Coreference	지시 대상 연결
Head 5-H	Task-specific patterns	태스크별 특수 패턴

Each head = independent Hopfield network with different Q, K, V projection matrices

Multi-Head Attention = H independent Hopfield networks searching different pattern subspaces simultaneously

Why does Transformer work so well?

$$\xi_{new} = X^V \cdot \text{softmax}(\beta (X^K)^T \xi^Q) \equiv \text{Attention}(Q, K, V)$$

Exponential Capacity

Attention can retrieve from $2^{(d/2)}$ patterns — virtually unlimited

1-Step Convergence

One forward pass through Attention = one complete Hopfield retrieval

Multi-Head = Parallel Search

H heads = H independent Hopfield networks exploring different subspaces

Storage Capacity Analysis

저장 용량 분석

"The new Hopfield network has exponential storage capacity." — Ramsauer et al.

Theorem 1: retrieval condition

Theorem 2: capacity

Error bound

Theorem 1: Retrieval Condition / 검색 조건

PART 5

Theorem 1 — Correct retrieval in one update step

$$\Delta_{\min} > \frac{1}{\beta} \ln(N-1) + \frac{M_{\max}^2}{2} - \frac{\Delta_{\min}^2}{2}$$

Notation / 기호 설명

$\Delta_{\min}$ minimum separation between stored patterns

$M_{\max}$ maximum norm of stored patterns

N number of stored patterns

β inverse temperature parameter

Interpretation / 해석

If stored patterns are sufficiently separated from each other ($\Delta_{\min}$ is large), a query ξ near one pattern will be correctly retrieved in a single update step.

패턴들 간 최소 거리가 충분히 크다면, 단 1번의 업데이트로 올바른 패턴 검색 가능

Theorem 1: sufficient separation between patterns guarantees one-step exact retrieval

Theorem 2: Exponential Storage Capacity / 지수적 저장 용량

PART 5

Theorem 2 — Exponential capacity of the modern Hopfield network

$$C_{\text{modern}} = 2^{\alpha d/2}$$

Model	Capacity	d=100	d=1000
Classical Hopfield	$0.14 \times d$	~ 14	~ 140
Dense (n=3)	$\sim d^2$	10,000	10^6
Dense (n=5)	$\sim d^4$	10^8	10^{12}
Modern Hopfield (this paper)	$2^{\alpha d/2}$	$2^{50} \approx 10^{15}$	$2^{500} \approx \infty$

Exponential capacity: d=1000 neurons $\rightarrow 2^{500}$ patterns — compared to only 140 in classical Hopfield

Theorem — Expected retrieval error

$$P(\text{error}) \leq e^{-\beta \Delta_{\min}^2 / 2}$$

What this means / 의미

When $\Delta_{\min}$ (pattern separation) is large relative to $1/\beta$, the probability of retrieval error decreases exponentially.

Pattern separation이 충분할 때, 검색 오류 확률은 지수적으로 작아짐.

Connection to Transformer / 트랜스포머와의 연결

In Transformer, K (keys) are learned representations of input tokens. When different tokens have distinct representations, $\Delta_{\min}$ is large, and attention retrieves accurately.

학습된 k (키) 벡터들이 잘 분리될수록 어텐션의 정확도가 지수적으로 향상

Convergence Properties

수렴 특성

"The iteration of the update rule converges globally, and typically converges in one update step." — Ramsauer et al.

Energy monotonicity

1-step convergence theorem

Theorem 3: Energy Decrease / 에너지 단조 감소

$$E(\xi_{\text{new}}) \leq E(\xi_{\text{old}})$$

Every single update step is guaranteed to decrease (or maintain) the energy. The network always converges — it cannot oscillate.

모든 업데이트는 에너지를 반드시 감소시킴 → 진동 없이 반드시 수렴

Theorem 4: One-Step Convergence / 1회 수렴

$$\Delta_{\min} \geq \sqrt{4\ln(N)/\beta} \Rightarrow \text{converges in } \mathbf{1} \text{ update}$$

When patterns are well-separated, a SINGLE update step is sufficient for exact retrieval. This directly explains why a single Attention layer works effectively.

패턴이 충분히 분리되면 단 1회 업데이트로 완벽 검색 → Attention 1번으로 충분한 이유

Convergence: Classical vs. Modern / 수렴 특성 비교

PART 6

Property	Classical Hopfield	Modern Hopfield (= Transformer Attention)
Update rule	$\text{sign}(Wx)$	$X \cdot \text{softmax}(\beta X^T \xi)$
Convergence guarantee	Always (global)	Always (global)
Typical steps to converge	Many iterations	Usually 1 step
Convergence speed	Slow (linear)	Fast (exponential)
Energy function	Quadratic $E = -\frac{1}{2}x^T Wx$	Is-e-based E
Stationary points	Stored patterns (+ spurious)	Stored patterns (clean)

Modern Hopfield's 1-step convergence is the theoretical basis for the single-pass efficiency of Transformer attention

Experiments & Applications

실험 및 응용

"We introduce a new layer — the Hopfield layer — which can be used as a drop-in replacement for the attention mechanism." — Ramsauer et al.

Hopfield Layer

Immune repertoire

BERT analysis

Hopfield Layer / 홉필드 레이어

A trainable neural network layer based on the modern Hopfield update rule.

Key properties:

- Plug into any existing architecture
- Learnable pattern matrices (X^K , X^V)
- Adjustable β per head
- Mathematically equivalent to attention

When to use Hopfield Layer / 활용 시나리오

Very large N

When the number of patterns to store is enormous (e.g. immune datasets)

β tuning

When different retrieval sharpness is needed across layers

Interpretability

When you want to analyze what patterns are being retrieved

Hybrid arch.

When combining with classical layers in novel architectures

Task / 태스크

Classify cancer patients vs. healthy controls from immune cell receptor sequences (antibody repertoires). Each patient has tens of thousands of sequences.

환자/정상 면역 서열로 암 분류 — 패턴 수가 극도로 많은 케이스

Why Hopfield excels here / 홉필드의 강점

Challenge: Millions of sequences — far beyond classical Hopfield capacity

Solution: Modern Hopfield's $2^{(\alpha d/2)}$ capacity handles the scale

Result: Hopfield Layer significantly outperforms standard Transformer attention on rare-pattern detection tasks

"Hopfield networks provide a framework to interpret what BERT's attention heads learn." — Ramsauer et al.

Small β heads

- Broad retrieval — attend to many patterns simultaneously
- Capture general semantic context
- $\beta \sim$ context-width trade-off

Large β heads

- Sharp retrieval — focus on single closest pattern
- Capture specific syntactic relations
- Converge quickly to one token

Layer depth trend

- Early layers: small β , broad attention
- Late layers: larger β , more focused
- Hierarchical abstraction explained by Hopfield theory

Conclusion

결론 및 요약

Connecting associative memory theory to modern deep learning architectures

Key contributions

Final answer

Summary of Key Contributions

PART 8

01	Continuous Generalization	Extended classical Hopfield to real-valued (continuous) states with a new energy function using log-sum-exp
02	Exponential Capacity Theorem	Proved that the modern Hopfield network stores $2^{(\alpha d/2)}$ patterns — exponential vs. classical linear $0.14d$
03	Equivalence Theorem	Proved that the modern Hopfield update rule is mathematically identical to Transformer scaled dot-product attention
04	One-Step Convergence	Proved that under mild separation conditions, exact retrieval is achieved in a single update (= one Attention layer)
05	Hopfield Layer	Introduced a drop-in neural network layer applicable to any architecture, with principled Hopfield interpretation

This paper bridges 30+ years of Hopfield network theory with state-of-the-art Transformer architectures

Why does Transformer work so well? / 왜 트랜스포머가 잘 작동하는가?

"Because Transformer Attention IS a Modern Hopfield Network performing pattern retrieval with exponential storage capacity — one single Attention operation achieves near-perfect retrieval."

$$\xi_{new} = X^V \cdot \text{softmax}(\beta (X^K)^T \xi^Q) \equiv \text{Attention}(Q, K, V)$$